

CO4 AT3 OUTPUTS

1.

```
er 60204 -- C:\Users\reddy\lnp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO4
Sentence: The student reads a book.

CFG Tree:
S
|-- NP
|   |-- The
|   |-- student
|-- VP
|   |-- reads
|   |-- NP
|       |-- a
|       |-- book

Dependency Structure:
reads -> student
student -> The
reads -> book
book -> a
```

2.

```
er 60204 -- C:\Users\reddy\lnp\DSA0304-NATURAL-LANGUAGE-PROCESSING\CO4
Input: Book a flight to Delhi

Top-Down Parsing:
Start -> Sentence
Sentence -> Verb + Object
May require backtracking

Earley Parsing:
Processes words from left to right
Stores partial results
Handles incomplete input
Handles ambiguity

Conclusion:
Earley parsing is better for dynamic input.
PS C:\Users\reddy\lnp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
```

3.

```
Sentence: She saw the man with a telescope.

CFG:
Parse 1: She used the telescope.
Parse 2: The man had the telescope.

PCFG:
Parse 1 probability = 0.70
Parse 2 probability = 0.30
Selected Parse 1

Neural Parser:
Uses context and learned language patterns
Can resolve complex ambiguity

Conclusion:
Neural parsing is suitable for real-world applications.
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
```

4.

```
er' '60701' '--' 'C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PR
Feature Structure:
{'word': 'student', 'number': 'singular'}
{'word': 'reads', 'number': 'singular'}
Subject-Verb Agreement: Correct

Subcategorization:
give -> Subject + Verb + Recipient + Object
The teacher gave the student a book.
Subject: teacher
Verb: gave
Recipient: student
Object: book
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
```

5.

```
er 60761 -- C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
Sentence: The student reads a book.

Transition-Based Parsing:
SHIFT -> The
SHIFT -> student
RIGHT-ARC -> student
SHIFT -> reads
RIGHT-ARC -> reads
SHIFT -> book
RIGHT-ARC -> book

Graph-Based Parsing:
Create possible dependency relationships
Compare possible structures
Select the best dependency tree

Comparison:
Transition-Based -> Fast and memory efficient
Graph-Based -> Considers global structure

Conclusion:
Transition-Based parsing is better for large-scale applications.
PS C:\Users\reddy\nlp\DSA0304-NATURAL-LANGUAGE-PROCESSING>
```